The Ford–Johnson algorithm, no longer unbeaten for less than 47 elements

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Abstract

It is shown that 26 elements can be sorted with 90 comparisons, one fewer than merge insertion requires.

Keywords: Analysis of algorithms, Sorting, Merge insertion, Formal verification

In 2007 Peczarski [1] found the algorithm of Ford and Johnson [2] 'still unbeaten for less than 47 elements'. We show that it no longer is.

Let $S(n)$ denote the least number of comparisons sufficient to sort n distinct elements. The algorithm of Ford and Johnson, now usually called merge insertion, sorts n elements with at most

$$F(n) = \sum_{k=1}^{n} \left\lceil \log_2 \frac{3k}{4} \right\rceil$$

comparisons [3], and it is known that $S(n) = F(n)$ for every $n \le 22$ [1, 3–5]. Manacher [6] proved that $S(n) < F(n)$ for infinitely many n; the least such n previously known was 47, by a result of Schulte Mönting [7].

Theorem. *$S(26) \le 90$ and $S(46) \le 194$, whereas $F(26) = 91$ and $F(46) = 195$. More generally, $S(n) < F(n)$ whenever $26 \le n \le 32$ or $46 \le n \le 64$.*

Proof. We sort 21 of the 26 elements by merge insertion, which requires $F(21) = 66$ comparisons. The remaining five have not yet been compared with anything, and they may be inserted into the ordered chain of 21 by means of a decision tree $T_{21,5}$ of depth 24, given in [8]. This makes 90 comparisons in all. In the same way, 42 of 46 elements may be sorted with $F(42) = 171$ comparisons and the remaining four inserted by a tree $T_{42,4}$ of depth 23, which makes 194. Finally, binary insertion gives $S(n+1) \le S(n) + \lceil \log_2(n+1) \rceil$, and for $26 \le n < 32$ and $46 \le n < 64$ this increment, five and six respectively, equals $F(n+1) - F(n)$. Hence $S(n) \le F(n) - 1$ implies $S(n+1) \le F(n+1) - 1$, and the improvement extends to 32 and 64 elements. □

Inserting the new elements one at a time by binary insertion would take 25 and 24 comparisons, and the information-theoretic lower bounds are 23 and 22; every leaf of either tree is reached by exactly one input. Since $26! > 2^{88}$, at least 89 comparisons are necessary, so $S(26)$ is either 89 or 90. For 47 elements the argument gives 200, the bound of Schulte Mönting. The proof has been formalised in Lean 4 [9], using `native_decide` to check the trees because of their size; the formalisation and the trees are available at [8]. They were found by computer search, the specific technique of which may be described elsewhere.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude Opus 5.5 (Anthropic) in order to assist in writing up this note. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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